

Madhusudhanan RAMAKRISHNAN

Immediate Joiner

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Profile Summary

- **Software Engineer** with 11+ years of experience in embedded systems, data engineering, mobile app development (Flutter & Android), and expertise in the automotive and rail industries. Adept at developing, testing, and delivering efficient, scalable solutions across diverse platforms.
- Version control with **Git**, testing, debugging, and performance tuning.
- Expertise in **System Integration Testing** for both Software-Software and Software-Hardware interactions.
- Proficient in **Agile and V-Model** methodologies, leveraging frameworks like Scrum and Kanban for effective project management.
- Extensive experience in **debugging, troubleshooting and validating** embedded applications.
- Skilled in conducting **end-to-end system validation**, with a strong grasp of SDLC, STLC, and software architecture principles.
- **Collaborative team player, adept at partnering with cross-functional teams** to enhance code quality and troubleshoot issues.
- Developed **automated testing frameworks** and **test management platforms**, achieving a 20% reduction in testing cycles.
- **Built and maintained cross-platform mobile applications using Flutter**, achieving high performance and a consistent experience across Android and iOS platforms.

Skills

Technical Proficiencies:

- **Programming Knowledge:** Visual C#, C++, .Net, VBA, C, Embedded C, Assembly Language, Verilog
- **Data Analysis:** Excel, Python, SQL, CSV, Apache Flink, AWS, Data Structures, PySpark, PowerBI
- **Standards and Compliance:** ISO 26262, EN 50128, EN 50126, SAE, ISO, IEC
- **Protocols:** UART, RS-232, RS-485, RS-422, CAN, I2C, SPI, MODBUS, TCP-IP
- **Soft Skills:** Effective Communication, Problem-solving, Team Collaboration, Adaptability, Attention to Detail, Time Management, Analytical, Risk Management

Industry Knowledge:

- **Metro Rail:** Rolling Stock, Signalling, Safety, TCMS, ETCS, CBTC, HVAC
- **Automotive:** Vehicle Dynamics, Infotainment, SIL, HIL, HVAC
- **Data Engineer:** ETL Process, Database Management, Real-time, NLP, Snowflake
- **Electronic Design:** Circuit Design, PCB, Oscilloscope, Wireshark, Analyzer, Emulation
- **Mobile Development:** Flutter, Android (Java, Kotlin, Dart)

Software & Tools:

- **Project Management:** Agile, Verification & Validation, Scrum, UVM
- **Management Tools:** JIRA, IBM Rational Team Concert, IBM DOORS, Bugzilla, GIT, ClearCase, ClearQuest, Team Foundation Server, PDM, GitHub
- **Testing Methodologies:** Manual Testing, Automated Testing, Regression Testing, Integration Testing, System Testing, Functional Testing, Automation
- **Environments:** Visual Studio, Keil, IAR, MATLAB, RQM
- **Platforms:** Windows, Linux, RTOS

Work History

Alstom Transport India Limited | Hyderabad, Telangana, India.
Senior Software Engineer | February 2023 to October 2024.

- Collaborate with cross-functional teams to monitor, troubleshoot, and integrate the applications such as embedded and data processing.
- Involved in ETL process and Apache Flink framework to process data from GPS sensors and other tracking systems to monitor train locations in real-time.
- Utilized Android Studio and Flutter DevTools for debugging, testing, and deploying apps for the organisation.
- Explored and participated in AI technologies and worked with SQL databases.
- Conducted rigorous validation and worked closely with developers to develop and refine the product, ensuring optimal performance, scalability, and user experience.
- Deliver test deliverables including Software Test Specifications, Traceability and Software Test Reports for Audit.
- Develop manual and automated test cases based on user requirements.
- Check, and review the requirements as well as the technical documents.
- Identify and examine the software bugs or defects.
- Develop, test, and maintain Python and C# code.
- Perform the review of the test deliverables as a part of License Checker and Peer Review.
- Participated in discussion on designs of functional and systems and in QA or code review.

QuEST Global Engineering Services Private Limited | Hyderabad, Telangana, India.
Lead Engineer | February 2021 to September 2022.

- Involved in SIL related test activities.
- Delegate and supervise tasks within the project team.
- Review, verify, and update the technical documents using PDM.
- Identified and reported potential compliance issues and worked with development team to resolve them.
- Handled and worked with both SQL and NoSQL databases.
- Experience testing and validating programs in command-line and IDE-based environments.
- Involved in the discussion on architecture level and in the activity of test deliverables.
- Experience with real-time data processing, and system integration.

Cyient Limited, Hyderabad, Telangana, India.
Senior Software Engineer | May 2017 to January 2021.

- Present weekly QA reports (i.e. identifying the number of bug fixes, new features pending/completed, etc.).
- Deliver Software Test Specifications and Software Test Reports for Audit.
- Installation of the software on the test racks and building Virtual Train.
- Develop C# code, debug, test and execute the test scripts for CCU-S Integration Testing.
- Performed system-level testing and debugging to identify and fix potential bugs.

Tristar Wireless Solutions Private Limited, Coimbatore, Tamil Nadu, India
Test Engineer | August 2013 to May 2017

- Development of test specifications for the high-level requirements for each function group and verify integration of function groups.
- Use of CSV files as scripts to automate tests and log results.
- Aided competitive vehicle driving events and benchmark tests.
- Performed integration tests on logistics-related software to optimize system performance, and to ensure reliability.
- Collaborated with design and backend teams to implement custom UIs, APIs.
- Hands-on experience in Emulation and involved in formal verification.

LAB Electronics, Chennai, Tamil Nadu, India
Lab Technician | May 2010 to March 2011

- Designing and troubleshooting of the circuits and modules.
- Testing and verification of the embedded devices and modules.
- Developing scripts based on the requirements.
- Exposure to System Verilog and VHDL under UVM methodology.
- Experience with hardware components on PCB and their analysis tools.

Education

- Master of Science in Applied Electronics, 2011-2013, PSG College of Arts & Science, Coimbatore, Tamil Nadu, India.
- Bachelor of Science in Electronics, 2007-2010, St. Joseph's College, Tiruchirappalli, Tamil Nadu, India.

Certifications

Certified in **ISTQB (International Software Testing Qualification Board)** foundation level conducted by ITB (Indian Testing Board).

Achievements

- I have been recognized as the "Most Valued Performer" three times.
- I have been awarded for my contribution to "ISA Audit Clearance" especially for quality.
- A Silver trophy was awarded for the adaptation of a new technique that reduces hours of time.

Languages

Telugu, Tamil, and English.